

How-To: ragent Configuration (ragent.json)

ragent is configured through a single JSON file called `ragent.json`. This document is the complete reference for every section of that file: what it controls, the schema, the defaults, and worked examples.

If you only want to get running quickly, read the **Quick Start** section in the project root `README.md` and the **Tutorial** in `docs/howtos/tutorial.md`. This document is the deep reference.

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1. Overview

All ragent runtime behaviour is driven by a single layered JSON configuration:

- **Provider connections** (API keys, base URLs, thinking settings, model catalogues)
- **Agent defaults and overrides** (system prompts, max steps, skills)
- **Permissions** (allow / deny / ask rules, bash and directory allow/deny lists)
- **Tool visibility** (which tool families are advertised to the LLM)
- **Memory, compaction, and retrieval** settings
- **Telemetry / OpenTelemetry export**
- **External integrations** (GitLab, Telegram/Discord channels, Gmail, finance data providers)
- **Feature flags** (experimental, SDD, pie-gap, triggers, research)

The config is loaded once at startup by `ragent-config::Config::load`, merged across layers, and then held in memory for the lifetime of the process. Several subsystems (YOLO mode, edit logging, bash/directory lists) also read runtime atomic flags that are synced from the config at startup.

Design principles

Principle	Meaning
Layered merge	Global config provides defaults; project config overrides; env vars override both.
Explicit-over-default	Many fields track whether the user <i>explicitly</i> set them so a serialise round-trip does not lose a toggle.
Sensible defaults	Every section has <code>Default</code> implementations so an empty <code>{}</code> is valid.
No secrets in config by force	API keys come from env vars or <code>env:VAR_NAME</code> indirection; the config file can reference them but does not have to store them.
Atomic persistence	Writes use temp-file-then-rename so a crash never leaves a partial config.

2. File Discovery and Loading Order

`Config::load` resolves configuration in a strict precedence order. Each layer is merged on top of the previous one, with later layers overriding earlier ones for explicitly-set fields.

compiled defaults

```
-> global file:  <config_dir>/ragent/ragent.json
-> project file:  ./ragent/ragent.json
-> env file:      $RAGENT_CONFIG (path to a JSON file)
-> env inline:    $RAGENT_CONFIG_CONTENT (raw JSON string)
```

Platform config directory

The global config directory is resolved by the `dirs` crate:

Platform	Path
Linux	<code>\$XDG_CONFIG_HOME/ragent/ragent.json</code> (default <code>~/.config/ragent/ragent.json</code>)
macOS	<code>~/Library/Application Support/ragent/ragent.json</code>
Windows	<code>%APPDATA%\ragent\ragent.json</code>

The helper `Config::global_config_dir()` returns `<config_dir>/ragent` and `Config::global_config_path()` returns the full `ragment.json` path.

Project config

The project config lives at `./ragent/ragent.json` relative to the current working directory. This is the file created automatically on first run when neither a global nor project config exists.

JSONC note

The crate-level documentation mentions `ragment.jsonc` as a supported filename, and JSONC-style comments appear in documentation examples (`//` line comments). However, the parser uses `serde_json::from_str`, which is strict JSON. Inline `//` comments are **not** supported in the actual config file. Use a `.json` extension and plain JSON. The `jsonc` examples in code doc-comments are for illustration only.

Auto-creation

If no config file is found at either the global or project path, `Config::load` creates a `.ragent/` directory and writes a default `ragment.json` there so the user has a starting point.

Load cache and save invalidation

`Config::load()` keeps a process-level cache of the resolved configuration, keyed on the current working directory plus the `(path, mtime, size)` tuple of every candidate config file that existed at load time. Two rules keep the cache coherent (the M-025 discipline):

- The mtime snapshot is taken **after** `load_uncached()` runs, so a first-time load that auto-creates `.ragent/ragent.json` includes the new file in the cache key.
- Every successful write through `Config::save()` or `Config::save_to_source()` calls `Config::invalidate_load_cache()`, so the next `load()` re-reads from disk instead of serving the pre-save config.

Together these guarantee that TUI saves (`/codeindex off`, `/tools`, ...) are immediately visible to subsequent loads in the same process — including the case where the save created a config file that was absent from the cached mtime snapshot (for example, a fresh environment with no global config, as on CI).

3. Merge Semantics

`Config::merge(base, overlay)` combines two configs. Different sections use different merge strategies:

Section	Strategy
username, API keys, gitlab, channels, gmail	Overlay overrides base when Some.
defaultAgent	Overlay overrides when explicitly set or different from default.
provider	Deep merge per provider key: <code>provider-level api</code> , <code>env</code> , <code>thinking</code> , and <code>options</code> are replaced when the overlay specifies them, while <code>models</code> entries are merged per-model-id so a partial overlay (e.g. <code>provider.openrouter.models."anthropic/claude-sonnet-4"</code>) does not discard lower-layer fields such as <code>name</code> or <code>capabilities</code> . Model-level <code>thinking</code> overrides provider-level <code>thinking</code> .
agent, command, mcp	Per-key overlay replaces base entry.
permission, instructions, skill_dirs, hooks	Append (union).
bash.allowlist, bash.denylist	Union (deduplicated).
prices	Per-model last-wins (overlay entry replaces base entry for the same model id).
code_index, tool_visibility	Overlay wins only for explicitly-set fields. The specified flags are propagated so the value survives a reload.
compaction	Overlay replaces base entirely.
yolo, edit_log	OR semantics: if the overlay sets <code>true</code> , it wins.
sdd, piegap	OR semantics: a flag enabled in either layer stays enabled.
research.open_access_recovery, research.exclude_academic_engines	OR semantics. <code>contact_email</code> , <code>oa_min_full_text_chars</code> , <code>max_concepts</code> , and <code>max_findings</code> : overlay overrides when present (and different from the default).

Section	Strategy
telemetry	If overlay enables telemetry, the whole overlay otel block replaces the base. Otherwise maps (resource_attributes, metrics) are unioned.
experimental	open_telemetry and parallel_tool_calls: OR semantics. Other fields: overlay defaults.

The config_paths field records every file that contributed to the final merged config, in load order.

4. CLI and Environment Overrides

--config <PATH> CLI flag

```
ragent --config /path/to/my-config.json run "hello"
```

When --config is supplied, the file at that path is loaded directly and bypasses the normal discovery order. Parse errors for this path include the file path, line, column, the problematic source line, and a caret marker.

Environment variables

Variable	Purpose
RAGENT_CONFIG	Path to a JSON config file, merged on top of project config.
RAGENT_CONFIG_CONTENT	Raw JSON string, merged last (highest precedence). Parse errors show the inline content.
Provider API keys	See section 7.3. Env vars always take precedence over config-file values.
TAVILY_API_KEY	Overrides tavily_api_key.
OPENALEX_EMAIL	Overrides openalex_email.
EXA_API_KEY	Overrides exa_api_key.
SERPER_API_KEY	Overrides serper_api_key.
GITLAB_TOKEN, GITLAB_URL, GITLAB_USERNAME	Override gitlab fields (highest priority).

Other CLI flags that affect config

Flag	Effect
--model <provider/model>	Overrides the resolved model.
--agent <name>	Overrides defaultAgent.
--yes / --no-prompt	Auto-approves all permissions (like YOLO but does not persist).

5. Config Error Reporting

When a config file fails to parse, ragent produces an actionable diagnostic:

```
Failed to parse config file '.ragent/ragent.json':  
Error at line 12, column 8:
```

```
Problematic line:  
  "defaultAgent: "coder"  
  ^
```

```
Parse error: expected `:` at line 12 column 8
```

This includes:

- The file path (or RAGENT_CONFIG_CONTENT for the inline env var).
- The line and column from the serde error.
- A separator line.
- The exact source line that caused the failure.
- A caret (^) positioned under the error column.
- The underlying serde message.

The same format is used for `--config`, `RAGENT_CONFIG`, and `RAGENT_CONFIG_CONTENT` errors.

6. Backup and Restore

`Config::backup_global_config()` snapshots the current global `ragent.json` into a timestamped file inside `<config_dir>/ragent/saves/`. The backup name uses `YYYY-MM-DD.HH-MM-SS` (hyphens in the time portion for Windows NTFS compatibility). If multiple saves happen in the same second, a counter is appended. The write is atomic (temp-file-then-rename).

`Config::restore_global_config()` restores a backup over the global `ragent.json`. It validates that:

- The backup file exists and is a regular file.
- The resolved destination is exactly `<config_dir>/ragent.json`.
- The backup content is valid JSON.

The restore is also atomic.

These are used by the `/config backup` and `/config restore` slash commands.

7. Section Reference

7.1 username

Field	Type	Default	Description
username	Option<String>	None	Display name of the user, used in prompts and logs.

```
{ "username": "alice" }
```

7.2 defaultAgent

Field	Type	Default	Description
defaultAgent (alias default_agent)	String	"general"	Name of the agent used when none is specified via --agent.

```
{ "defaultAgent": "coder" }
```

The loader tracks whether the user explicitly set this field (`specified_default_agent`) so merge operations do not accidentally overwrite an explicit project-level choice with a global default.

7.3 provider

The provider section is the most important part of the config. It is a map keyed by provider id, where each entry configures one LLM provider.

```
{
  "provider": {
    "anthropic": {
      "env": ["ANTHROPIC_API_KEY"],
      "thinking": { "enabled": true, "level": "low" },
      "models": {
        "claude-sonnet-4-20250514": {
          "thinking": { "enabled": true, "level": "high", "budget_tokens": 16000 }
        }
      }
    },
    "generic_openai": {
      "env": ["GENERIC_OPENAI_API_KEY"],
      "api": { "base_url": "http://127.0.0.1:8080" }
    }
  }
}
```

ProviderConfig schema

Field	Type	Default	Description
env	Vec<String>	[]	Environment variable names required by this provider. ragent checks these are set before model discovery.
api	Option<ApiConfig>	None	Optional API endpoint and header overrides.

Field	Type	Default	Description
thinking	Option<ThinkingConfig>	None	Default thinking/reasoning config for all models under this provider.
models	HashMap<String, ModelConfig>		Overridden by per-model thinking. Model definitions keyed by model id.
options	HashMap<String, Value>		Arbitrary provider-specific options.

ApiConfig schema

Field	Type	Default	Description
base_url	Option<String>	None	Base URL for API requests, overriding the provider default.
headers	HashMap<String, String>		Extra HTTP headers sent with every request.

ModelConfig schema

Field	Type	Default	Description
name	Option<String>	None	Human-readable display name.
cost	Option<Cost>	None	Per-token pricing (USD per million tokens).
capabilities	Option<Capabilities>	None	Feature flags for this model.
thinking	Option<ThinkingConfig>	None	Per-model thinking config; overrides provider-level default.

Cost schema

Field	Type	Description
input	f64	Cost per million input tokens (USD).
output	f64	Cost per million output tokens (USD).

Capabilities schema

Field	Type	Default	Description
reasoning	bool	false	Supports chain-of-thought reasoning.
streaming	bool	true	Supports streaming responses.
vision	bool	false	Can process image inputs.
tool_use	bool	true	Supports tool/function calling.
thinking_levels	Vec<ThinkingLevel>		Which thinking/reasoning levels this model supports.

ThinkingConfig schema Defined in `ragent-types`, this is shared across providers.

Field	Type	Default	Description
<code>enabled</code>	<code>bool</code>	<code>true</code>	Whether thinking is enabled. When <code>false</code> , level is ignored.
<code>level</code>	<code>ThinkingLevel</code>	<code>auto</code>	Thinking depth: <code>auto</code> , <code>off</code> , <code>low</code> , <code>medium</code> , <code>high</code> .
<code>budget_tokens</code>	<code>Option<u32></code>	<code>None</code>	Max tokens for thinking (Anthropic <code>budget_tokens</code>). <code>None</code> = provider default.
<code>display</code>	<code>Option<ThinkingDisplay></code>	<code>None</code>	How thinking content is surfaced: <code>full</code> , <code>summarized</code> , <code>omitted</code> .

`ThinkingLevel` maps to provider-native parameters:

Provider	Native parameter
Anthropic	<code>thinking.type + effort / budget_tokens</code>
OpenAI / Copilot	<code>reasoning_effort</code> (<code>low</code> , <code>medium</code> , <code>high</code> , <code>none</code>)
Gemini	<code>thinkingConfig.thinkingLevel</code> (<code>minimal</code> , <code>low</code> , <code>medium</code> , <code>high</code> , <code>auto</code>)
Ollama	<code>think</code> boolean

Supported provider ids Each provider is implemented in `crates/ragent-llm/src/providers/`. The provider id (from `fn id() -> &'static str`) is the key in the provider map.

Provider id	Env var(s)	Notes
<code>anthropic</code>	<code>ANTHROPIC_API_KEY</code>	Claude models. Supports <code>thinking/budget_tokens</code> .
<code>openai</code>	<code>OPENAI_API_KEY</code>	GPT models. Supports <code>reasoning_effort</code> .
<code>openai_responses</code>	<code>OPENAI_API_KEY</code>	OpenAI Responses API variant.
<code>gemini</code>	<code>GOOGLE_API_KEY</code> or <code>GEMINI_API_KEY</code>	Google Gemini. Supports <code>thinkingConfig</code> .
<code>ollama</code>	<code>OLLAMA_API_KEY</code> (optional for local)	Local Ollama. <code>think</code> boolean for thinking.
<code>ollama_cloud</code>	<code>OLLAMA_API_KEY</code>	Ollama Cloud variant.
<code>huggingface</code>	<code>HF_TOKEN</code> or <code>HUGGING_FACE_HUB_TOKEN</code>	Hugging Face Inference.

Provider id	Env var(s)	Notes
copilot	GITHUB_COPILOT_TOKEN or GITHUB_TOKEN (via gh auth)	GitHub Copilot.
generic_openai	GENERIC_OPENAI_API_KEY or OPENAI_API_KEY; GENERIC_OPENAI_API_BASE for base URL	Any OpenAI-compatible endpoint.
azure_foundry	AZURE_AI_FOUNDRY_API_KEY; AZURE_AI_FOUNDRY_BASE for base URL	Azure AI Foundry.
azure_resource	(from azureresources.json)	Azure Resource (File) provider. Reads model definitions from a separate azureresources.json file.
bedrock	AWS_ACCESS_KEY_ID, AWS_SECRET_ACCESS_KEY, AWS_REGION / AWS_BEDROCK_REGION, optional AWS_SESSION_TOKEN	Amazon Bedrock. Uses AWS SigV4.
xai	XAI_API_KEY; XAI_API_BASE for base URL	xAI (Grok).
router	(aggregates other providers)	Model Router virtual provider. Classifies prompts into tiers and routes to the best model.

Model Router configuration The router is configured under `provider.router` and has its own RouterConfig:

```
{
  "provider": {
    "router": {
      "enabled": true,
      "tiers": {
        "SIMPLE": { "models": [{ "provider": "ollama", "model": "llama3.2" }] },
        "MEDIUM": { "models": [{ "provider": "anthropic", "model": "claude-sonnet-4-20250514" }] },
        "COMPLEX": { "models": [{ "provider": "anthropic", "model": "claude-sonnet-4-20250514" }] },
        "REASONING": { "models": [{ "provider": "openai", "model": "o3" }] }
      },
      "boundaries": { "simple_medium": 0.25, "medium_complex": 0.50, "complex_reasoning": 0.75 },
      "context_messages": 3,
      "default_timeout_ms": 30000
    }
  }
}
```

```

    }
}

```

Field	Type	Default	Description
enabled	bool	false	Master switch. When false, requests pass through to the MEDIUM tier default without classification.
tiers	HashMap<String, TierConfig>		Per-tier model fallback chains. Keys: SIMPLE, MEDIUM, COMPLEX, REASONING.
weights	WeightConfig (15 dimensions, summing to 1.0)		Classifier dimension weights.
boundaries	BoundaryConfig	0.25 / 0.50 / 0.75	Score boundaries between tiers. Must be ascending and in [0.0, 1.0].
context_message_size	usize	3	Number of recent conversation messages included in classification.
default_timeout_ms	ms	30000	Default request timeout.

TierConfig has models: Vec<TierEntry> (ordered fallback list) and optional timeout_ms. TierEntry has provider and model string fields.

The WeightConfig has 15 dimension weights (token_count, vocabulary_complexity, syntax_complexity, domain_specificity, ambiguity, context_dependency, reasoning_depth, creativity_level, emotional_complexity, multimodality, instruction_complexity, knowledge_recency, code_complexity, mathematical_complexity, image_attachment). Weights are normalised to sum to 1.0 at load time.

For more on providers and models, see the Tutorial (docs/howtos/tutorial.md).

7.4 agent

Per-agent configuration overrides, keyed by agent name. Applied on top of built-in agent defaults.

```

{
  "agent": {
    "coder": {
      "model": "anthropic/claude-sonnet-4-20250514",
      "temperature": 0.2,
      "max_steps": 200,
      "skills": ["rust-review"],
      "permission": [
        { "permission": "file:write", "pattern": "src/**", "action": "allow" }
      ]
    }
  }
}

```

AgentConfig schema

Field	Type	Default	Description
model	Option<String>	None	Model in "provider:model" format.
variant	Option<String>	None	Agent variant selector.
prompt	Option<String>	None	System prompt override.
temperature	Option<f32>	None	Sampling temperature override.
top_p	Option<f32>	None	Top-p (nucleus) sampling override.
mode	Option<String>	None	Agent mode: "primary", "subagent", or "all".
hidden	bool	false	Hide from user-facing listings.
permission	Vec<PermissionRule>		Permission rules specific to this agent.
max_steps	Option<u32>	None	Maximum agentic loop iterations. Per-agent override of the loop step budget; when unset the loop.max_steps default (512) applies.
skills	Vec<String>	[]	Skill names to preload into the prompt.
options	HashMap<String, Value>		Arbitrary agent-specific options.

For custom agents defined as JSON/Markdown files (OASF format), see `docs/howtos/custom-agents.md`.

7.5 permission

Global permission rules applied to all agents. Rules are evaluated last-match-wins using glob patterns.

```
{
  "permission": [
    { "permission": "file:write", "pattern": "src/**", "action": "allow" },
    { "permission": "file:write", "pattern": "secrets/**", "action": "deny" },
    { "permission": "bash", "pattern": "git push --force", "action": "deny" },
    { "permission": "*", "pattern": "/etc/**", "action": "deny" }
  ]
}
```

PermissionRule schema

Field	Type	Description
permission	String (parsed to Permission)	The permission type. Supports flat names (read, bash) and namespaced categories (file:read, file:write).
pattern	Option<String>	Glob pattern matched against the resource path. None matches all.
action	PermissionAction	allow, deny, or ask.

Standard permission types

Name	Permission variant	Description
read / file:read	Read	Read access to files/resources.
edit / write / file:write	Edit	Write/edit access.
bash / execute	Bash	Shell command execution.
web / fetch	Web	Network/web access.
question	Question	Interactive question to user.
plan_enter /plan	PlanEnter	Enter planning phase.
plan_exit	PlanExit	Exit planning phase.
task	Task	Create/modify task items.
external_dir	ExternalDirectory	Access dirs outside project root.
doom_loop	DoomLoop	Detect/break infinite loops.
*	Custom("*")	Wildcard: applies to all permissions.
any other string	Custom(String)	User-defined permission type.

Evaluation order

1. **Permanent grants** (runtime “always allow” recorded via `PermissionChecker::record_always`) are checked first.
2. **Permission-specific rules** are evaluated last-match-wins.
3. **Wildcard rules** (permission: `"*"`) are evaluated last.
4. If no rule matches, the result is ask.

For the full permission system, YOLO mode, and autopilot, see `docs/howtos/permissions.md`.

7.6 command

User-defined slash-command shortcuts. Each entry maps a slash command name to a shell command.

```
{
  "command": {
    "build": { "command": "cargo build --release", "description": "Build release binary" },
    "test": { "command": "cargo test -- --nocapture", "description": "Run all tests with output" }
  }
}
```

CommandDef schema

Field	Type	Description
command	String	Shell command to execute.
description	String	Human-readable description shown in help output.

7.7 mcp

MCP (Model Context Protocol) server definitions, keyed by server id.

```
{
  "mcp": {
    "filesystem": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "/home/user/docs"],
      "env": {}
    },
    "remote-api": {
      "type": "sse",
      "url": "https://mcp.example.com/sse",
      "headers": { "Authorization": "Bearer token123" },
      "disabled": false,
      "notification": "inject_summary"
    }
  }
}
```

McpServerConfig schema

Field	Type	Default	Description
type	McpTransport	stdio	Transport: stdio, sse, or http.
command	Option<String>	None	Executable path/name (stdio).
args	Vec<String>	[]	Command-line arguments (stdio).
env	HashMap<String, String>		Environment variables (stdio).
url	Option<String>	None	URL endpoint (SSE/HTTP).
headers	HashMap<String, String>		HTTP headers (SSE/HTTP).
disabled	bool	false	If true, server is configured but not started.
notification	McpNotificationMode	None	Push notification handling: none, inject_summary, inject_and_run.

7.8 instructions

Additional instruction strings appended to agent system prompts.

```
{
  "instructions": [
    "Always use rustfmt before writing Rust files.",
    "Prefer iterator adapters over manual loops."
  ]
}
```

Type: `Vec<String>`. Default: `[]`. These are appended to every agent's system prompt, merged (union) across config layers.

7.9 skill_dirs

Additional directories to scan for skill definitions, beyond the built-in locations.

```
{
  "skill_dirs": ["~/my-skills", "/shared/team-skills"]
}
```

Type: `Vec<String>`. Default: `[]`. Merged (union) across config layers.

7.10 experimental

Feature flags for experimental functionality.

```
{
  "experimental": {
    "open_telemetry": false,
    "parallel_tool_calls": false,
    "max_background_agents": 8,
    "background_agent_timeout": 3600
  }
}
```

ExperimentalFlags schema

Field	Type	Default	Description
open_telemetry	bool	false	Enable OpenTelemetry trace export (legacy; prefer <code>telemetry.otel</code>).
parallel_tool_calls	bool	false	Allow multiple tool calls from a single model turn to execute in parallel.
max_background_agents	usize	8	Maximum concurrent background sub-agent tasks.
background_agent_timeout	Duration	3600	Timeout in seconds for background sub-agent tasks.

`open_telemetry` is mapped as a deprecated alias: if `telemetry.otel` is not explicitly enabled but `experimental.open_telemetry` is true, telemetry is enabled with default settings and a deprecation warning is emitted.

7.11 hooks

Lifecycle hooks. Currently a placeholder (`Vec<serde_json::Value>`). See `docs/howtos/hooks.md` for the hooks system documentation.

Type: `Vec<Value>`. Default: `[]`. Merged (append) across config layers.

7.12 bash

User-defined bash command allowlist and denylist additions.

```
{
  "bash": {
    "allowlist": ["curl", "wget"],
    "denylist": ["git push --force", "rm -rf /"]
  }
}
```

BashConfig schema

Field	Type	Default	Description
allowlist	<code>Vec<String></code>	<code>[]</code>	Command prefixes exempted from the banned-command check (e.g. "curl").
denylist	<code>Vec<String></code>	<code>[]</code>	Substring patterns that always reject a command (e.g. "git push --force").

Both lists are unioned (deduplicated) across global and project configs. The `/bash add` and `/bash remove` slash commands mutate these lists and persist changes to the targeted config file.

Allowlist entries match the first word of the command. **Denylist** entries are substring-matched anywhere in the command string.

This interacts with the 7-layer bash security model. See `docs/howtos/permissions.md` for the full model.

7.13 dirs

User-defined directory/file path allowlist and denylist additions.


```

{
  "dirs": {
    "allowlist": ["src/**", "tests/**", "*.rs"],
    "denylist": ["secrets/**", "/etc/**", ".env"]
  }
}

```

DirsConfig schema

Field	Type	Default	Description
allowlist	Vec<String>	[]	Glob patterns for paths automatically allowed (no prompt).
denylist	Vec<String>	[]	Glob patterns unconditionally rejected.
allowed_roots	Vec<String>	[]	Extra directory roots permitted for path-escape checking (multi-root / sub-agent access); canonicalized at load.

Both lists are unioned across config layers. **Denylist takes precedence over allowlist.** The `/dirs add` and `/dirs remove slash` commands mutate these lists.

Built-in denylist patterns (always active, not configurable) include system directories: `/bin/**`, `/sbin/**`, `/boot/**`, `/dev/**`, `/proc/**`, `/sys/**`, `/etc/**`, `/usr/bin/**`, `/usr/sbin/**`, `/usr/lib/**`, `/lib/**`, `/lib64/**`, and their macOS/Windows equivalents.

7.14 tool_visibility

Controls which tool families are advertised to the LLM. When a switch is `false`, all tools in that family are hidden from tool definitions and system-prompt listings. Hidden tools remain registered and executable.

```

{
  "tool_visibility": {
    "office": true,
    "github": true,
    "gitlab": false,
    "teams": true,
    "agents": true,
    "plan": true,
    "codeindex": true,
    "masterfetch": true,
    "browser": true,
    "finance": true
  }
}

```

Switches and defaults

Switch	Default	Tools governed
office	false	office_read, office_write, office_info, libre_read, libre_write, libre_info, pdf_read, pdf_write
github	false	github_list_issues, github_get_issue, github_create_issue, github_comment_issue, github_close_issue, github_list_prs, github_get_pr, github_create_pr, github_merge_pr, github_review_pr
gitlab	false	19 GitLab tools (issues, MRs, pipelines, jobs)
teams	false	20 team coordination tools
agents	false	cancel_agent, list_agents, new_agent, agent_complete, wait_agents
plan	false	plan_enter, plan_exit
codeindex	true	codeindex_search, codeindex_status, codeindex_symbols, codeindex_references, codeindex_dependencies, codeindex_reindex
masterfetch	true	mf_fetch, mf_crawl, mf_search, mf_screenshot, mf_cache_clear, mf_version
browser	true	browser
finance	true	stock_quote, stock_history, stock_fundamentals, currency_rate, currency_history, stock_search, stock_options

The codeindex, masterfetch, browser, and finance switches are serialised **only** when explicitly set (tracked by specified flags). This lets the default config omit the key so code-level default changes propagate, while ensuring an explicit user toggle persists on save.

The /tools slash command toggles these at runtime. See docs/howtos/tool-visibility.md and docs/howtos/tools.md.

7.15 code_index

Persistent configuration for the codebase indexing subsystem.

```
{
  "code_index": {
    "enabled": true,
    "max_file_size": 1048576,
    "extra_exclude_dirs": ["vendor", "node_modules"],
    "extra_exclude_patterns": ["*.generated.rs"]
  }
}
```

CodeIndexConfig schema

Field	Type	Default	Description
enabled	bool	true	Whether code indexing is enabled.
max_file_size	u64	1048576 (1 MB)	Maximum file size in bytes to index.
extra_exclude_dirs	Vec<String>	[]	Additional directory names to exclude from scanning.
extra_exclude_patterns	Vec<String>	[]	Additional glob patterns to exclude.

The enabled field is serialised only when explicitly set (tracked by `specified.enabled`), so `/codeindex on`/`/codeindex off` toggles persist on save.

For the full code index guide, see `docs/howtos/codeindex.md`.

7.16 stream

LLM streaming behaviour: timeouts and retries.

```
{
  "stream": {
    "initial_response_timeout_secs": 300,
    "timeout_secs": 120,
    "max_retries": 4,
    "retry_backoff_secs": 2
  }
}
```

StreamConfig schema

Field	Type	Default	Validation	Description
initial_response_timeout_secs	u64	300	$\geq 5, \geq \text{timeout_secs}$	Seconds the HTTP client waits for the first byte of a streaming response.
timeout_secs	u64	120	≥ 5	Seconds of silence between stream deltas before a stream is considered stalled.
max_retries	u32	4	≤ 32	Maximum retry attempts after a stall or connection failure.
retry_backoff_secs	u64	2	—	Backoff multiplier per retry. Attempt N waits $N * \text{retry_backoff_secs}$ seconds.

`validate()` returns a list of problems; the session processor refuses to start if any are present.

7.17 memory

Memory system configuration: blocks, structured store, semantic search, retrieval, auto-extraction, decay, and cross-project sharing.

```
{
  "memory": {
    "enabled": true,
    "tier": "core",
    "structured": { "enabled": true },
    "semantic": {
      "enabled": false,
      "model": "all-MiniLM-L6-v2",
      "dimensions": 384
    },
  },
  "retrieval": {
    "max_memories_per_prompt": 100,
    "max_memory_tokens": 4000,
    "recency_weight": 0.3,
    "relevance_weight": 0.7
  },
  "auto_extract": {
    "enabled": false,
    "require_confirmation": true
  },
  "decay": {
    "factor": 0.95,
    "min_confidence": 0.1
  },
  "cross_project": {
    "enabled": false,
    "search_global": true,
    "project_override": true
  }
}
```

MemoryConfig schema

Field	Type	Default	Description
enabled	bool	true	Master switch for the memory system.
tier	String	"core"	Memory tier: "core" (file blocks), "structured" (SQLite store), "semantic" (with embeddings).
structured	StructuredMemoryConfig	enabled: true }	Structured store configuration.
retrieval	RetrievalConfig	(see below)	Retrieval configuration for prompt injection.

Field	Type	Default	Description
semantic	SemanticConfig	(see below)	Semantic search (embedding) configuration.
auto_extract	AutoExtractConfig	(see below)	Automatic memory extraction.
decay	DecayConfig	(see below)	Confidence decay.
cross_project	CrossProjectConfig	(see below)	Cross-project memory sharing.

Sub-schemas StructuredMemoryConfig: enabled: bool (default true).

RetrievalConfig:

Field	Type	Default	Description
max_memories_per_prompt	usize	100	Maximum structured memories fetched from SQLite for the system prompt. Acts as a safety cap.
max_memory_tokens	Option<usize>	4000	Approximate token budget for the injected memory section. null disables the budget (uses only the row cap).
recency_weight	f64	0.3	Weight for recency when ranking (0.0-1.0).
relevance_weight	f64	0.7	Weight for relevance when ranking (0.0-1.0).

SemanticConfig:

Field	Type	Default	Description
enabled	bool	false	Whether semantic search via embeddings is enabled. Requires the embeddings Cargo feature.
model	String	"all-MiniLM-L6-v2"	ONNX sentence-transformer model name.
dimensions	usize	384	Embedding vector dimensions (must match model output).

AutoExtractConfig:

Field	Type	Default	Description
enabled	bool	false	Whether automatic memory extraction is enabled.
require_confirmation	bool	true	Whether extracted candidates require explicit confirmation before storage.

DecayConfig:

Field	Type	Default	Description
factor	f64	0.95	Multiplicative decay factor per day since last access. 1.0 disables decay.
min_confidence	f64	0.1	Minimum confidence floor. Memories never decay below this.

CrossProjectConfig:

Field	Type	Default	Description
enabled	bool	false	Whether cross-project memory sharing is enabled.
search_global	bool	true	Whether search includes global memories.
project_override	bool	true	Whether project-specific blocks override global blocks with the same label.

7.18 compaction

Context-window compaction configuration (OpenCode-derived summarisation).

```
{
  "compaction": {
    "auto": true,
    "threshold": 0.7,
    "buffer": 0.10,
    "keep": { "tokens": 0.20 },
    "model": { "provider_id": "ollama", "model_id": "qwen2.5:1.5b" },
    "summary_tokens": 1500,
    "tool_output_max_chars": 2000
  }
}
```

CompactionConfig schema

Field	Type	Default	Description
auto	bool	true	Whether automatic pre-send compaction is enabled. When false, only provider context-overflow errors trigger emergency compaction.

Field	Type	Default	Description
threshold	Option<f64>	Some(0.7)	Fraction of context window at which to trigger compaction (0.0-1.0). None falls back to buffer-based trigger. Clamped to minimum 0.7 so routine prompts under 70% never trigger compaction.
buffer	f64	0.10	Token buffer as a fraction of context window. When threshold is None, triggers when tokens exceed $\text{context_window} - \max(\text{output_tokens}, \text{context_window} * \text{buffer})$.
keep	KeepConfig	{ tokens: Some(0.20) }	Recent turns kept verbatim after compaction.
model	Option<CompactModelRef>	None	Optional fast/cheap model override (provider_id + model_id) for the /compact summarisation call. The session's primary model is used when unset, and the pre-send auto-compaction path always uses the session's primary model (an AgentNotice is published when an override is configured).
summary_tokens	usize	1500	Maximum tokens requested for the compaction summary output.
tool_output_max_chars	usize	2000	Truncation limit for each tool output serialised into the summarisation prompt.

KeepConfig: tokens: Option<f64> (default Some(0.20)) — max fraction of context window preserved as the verbatim tail.

Runtime behaviour: - The summarisation prompt is capped adaptively at $\min(60,000 \text{ chars}, \text{context_window} * 4 / 2)$ so small local models get a proportionally smaller prompt. - The compaction stream has a hard 180-second overall cap plus a 60-second per-chunk stall timeout; progress is reported every 10 seconds with the elapsed time and the summary character count received so far. - The compaction agent's configured temperature is honoured by the summarisation request. - Cancellation is checked between stream chunks, so /compact honours the session cancel flag during the LLM call. - Token estimation uses an allocation-free byte counter instead of re-serialising tool JSON schemas.

7.19 gitlab

GitLab integration configuration. Values set here override those stored in the ragent database (set via /gitlab setup). Environment variables take the highest priority.

```
{
  "gitlab": {
    "instance_url": "https://gitlab.example.com",
    "token": "glpat-xxxxxxxxxxxx",
    "username": "myuser"
  }
}
```

GitLabIntegrationConfig schema

Field	Type	Default	Description
instance_url	Option<StringNone		GitLab instance base URL.
token	Option<StringNone		Personal Access Token.
username	Option<StringNone		GitLab username / identity.

Env var overrides: GITLAB_TOKEN, GITLAB_URL, GITLAB_USERNAME.

7.20 hidden_tools

Tool names to hide from the LLM. Hidden tools remain registered and executable; they are simply not advertised to the model.

```
{
  "hidden_tools": ["github_list_issues", "github_get_issue", "gitlab_list_mrs"]
}
```

Type: Vec<String>. Default: []. Unioned (deduplicated) across config layers.

The `effective_hidden_tools()` method also computes tools hidden by `tool_visibility` switches set to `false`.

7.21 yolo

YOLO mode bypasses all command validation and tool restrictions.

```
{ "yolo": false }
```

Type: bool. Default: `false`.

When enabled, the following safety checks are skipped: - Bash denied patterns (destructive commands like `rm -rf /` are allowed) - Dynamic context allowlist (any executable can run in skill bodies) - MCP config validation (shell metacharacters and unvalidated paths permitted)

Warning: This is inherently dangerous. Use only when you trust the agent and its inputs completely, or for local development/debugging.

The `Alt+Y` keybinding and `/yolo` slash command toggle this at runtime and persist the change. See `docs/howtos/permissions.md`.

7.22 edit_log

Edit-operation logging for edit and multi_edit tools.

```
{ "edit_log": false }
```

Type: bool. Default: false.

When enabled, all edit operations are logged. The Alt+E keybinding and /editlog slash command toggle this at runtime and persist the change.

7.23 prices

User-defined price overrides for cost estimation. Each entry overrides the built-in price table for a specific model. Prices are in USD per 1,000,000 tokens.

```
{
  "prices": [
    { "model": "gpt-4o", "input_per_1m": 2.50, "output_per_1m": 10.00 },
    { "model": "claude-sonnet-4-20250514", "input_per_1m": 3.00, "output_per_1m": 15.00 }
  ]
}
```

PriceEntry schema

Field	Type	Description
model	String	Model identifier as returned by the provider.
input_per_1m	f64	Price per 1M input/prompt tokens (USD).
output_per_1m	f64	Price per 1M output/completion tokens (USD).

When merged, overlay entries replace base entries with the same model id (last-wins per model).

7.24 browser

Browser automation configuration (Chrome DevTools Protocol endpoint).

```
{
  "browser": {
    "cdp_endpoint": "http://127.0.0.1:9222",
    "default_headless": true
  }
}
```

BrowserConfig schema

Field	Type	Default	Description
cdp_endpoint	Option<String>	None	CDP HTTP endpoint URL. When None/empty, defaults to <code>http://127.0.0.1:9222</code> .
default_headless	bool	true	Default headless mode for the setup action.

7.25 channels

External messaging channel configuration for the `send_channel_message` tool.

```
{
  "channels": {
    "enabled": true,
    "telegram": { "bot_token": "env:TELEGRAM_BOT_TOKEN", "chat_id": "-100123" },
    "discord": { "webhook_url": "https://discord.com/api/webhooks/..." }
  }
}
```

ChannelsConfig schema

Field	Type	Default	Description
enabled	bool	false	Master switch for the channel messaging tool.
telegram	Option<TelegramChannelConfig>	None	Telegram bot channel.
discord	Option<DiscordChannelConfig>	None	Discord webhook channel.

TelegramChannelConfig: `bot_token: Option<String>`, `chat_id: Option<String>`, `base_url: Option<String>` (defaults to `https://api.telegram.org`).

DiscordChannelConfig: `webhook_url: Option<String>`.

All token/secret fields support the `env:VAR_NAME` indirection — the value is read from the named environment variable at use time, so secrets do not need to live in the config file.

7.26 gmail

Gmail tool configuration (OAuth2 client credentials).

```
{
  "gmail": {
    "client_id": "...apps.googleusercontent.com",
    "client_secret": "env:GMAIL_CLIENT_SECRET"
  }
}
```

GmailConfig schema

Field	Type	Default	Description
client_id	Option<String>	None	OAuth2 client ID. Supports env:VAR_NAME. Falls back to GMAIL_CLIENT_ID.
client_secret	Option<String>	None	OAuth2 client secret. Supports env:VAR_NAME. Falls back to GMAIL_CLIENT_SECRET.
base_url	Option<String>	None	Optional HTTP endpoint override (defaults to https://gmail.googleapis.com).

The OAuth2 access/refresh tokens are managed by the gmail tool itself (auth/status/logout actions) and stored encrypted in ragent-storage — never in this file.

7.27 telemetry

OpenTelemetry metrics export configuration.

```
{
  "telemetry": {
    "otel": {
      "enabled": false,
      "endpoint": "http://localhost:4318",
      "protocol": "http",
      "export_interval_seconds": 30,
      "export_timeout_seconds": 10,
      "service_name": "ragent",
      "resource_attributes": { "deployment.environment": "production" },
      "metrics": { "ragent.tool.call_count": false },
      "internal_port": null,
      "cardinality_limit": 1000
    }
  }
}
```

TelemetryConfig schema Wraps OtelConfig as telemetry.otel.

OtelConfig schema

Field	Type	Default	Description
enabled	bool	false	Master on/off switch.
endpoint	String	"http://localhost:4318"	OTLP endpoint base URL. Must be http:// or https://.

Field	Type	Default	Description
protocol	OtelProtocol	http	Transport: http or grpc.
export_interval_seconds	int	30	Batch export interval. Must be > 0 when enabled.
export_timeout_seconds	int	10	Per-export request timeout. Must be > 0 when enabled.
service_name	String	"ragent"	service.name resource attribute. Must not be empty.
resource_attributes	HashMap<String, String>		Custom resource attributes appended to every metric.
metrics	HashMap<String, bool>		Per-metric enable/disable toggles. Absent = enabled by default.
internal_port	Option<u16>	None	Optional in-process Prometheus text endpoint port.
cardinality_limit_size	int	1000	Max distinct attribute combinations per metric before overflow into unknown bucket.

The legacy `experimental.open_telemetry` flag is mapped as a deprecated alias via `apply_legacy_flag()`.

7.28 agent_perf

Agent-loop performance configuration.

```
{
  "agent_perf": {
    "enabled": true,
    "profiling": false,
    "step_budget_secs": 300,
    "stall_timeout_secs": 60,
    "max_concurrent_tools": 4,
    "parallel_independent_tools": true
  }
}
```

AgentPerfConfig schema

Field	Type	Default	Validation	Description
enabled	bool	true	—	Master switch. When false, all performance optimisations short-circuit.
profiling	bool	false	—	Emit detailed per-scope timing logs at info level.
step_budget_secs	int	300	>= 5	Maximum wall-clock seconds per agent step.

Field	Type	Default	Validation	Description
stall_timeout	int64secs	60	>= 5	Maximum seconds without a stream delta before stall recovery fires.
max_concurrent_tools	int32tools	min(available_parallelism, 4)		Maximum parallel tool calls per turn.
parallel_independent_tools	bool		—	Execute independent tool calls in parallel.

validate() returns a list of problems; the agent loop refuses to start if any are present.

7.29 finance

Paid finance-provider configuration for the stock/currency toolset.

```
{
  "finance": {
    "provider": "alpha_vantage",
    "api_key": "env:ALPHA_VANTAGE_API_KEY",
    "requests_per_minute": 5,
    "min_call_interval_seconds": 5,
    "yahoo_fallback": true
  }
}
```

FinanceProviderConfig schema

Field	Type	Default	Description
provider	String	"yahoo"	Selected provider: "yahoo" (free, default) or "alpha_vantage".
api_key	Option<String>	None	API key when a paid provider is selected.
base_url	Option<String>	None	Optional base URL override for the paid provider.
requests_per_minute	Option<u32>	None	Optional request rate limit.
user_agent	Option<String>	None	Optional custom User-Agent header for the free Yahoo provider.
min_call_interval	int64seconds	5	Minimum seconds between any two finance API calls.
yahoo_fallback	Option<bool>	None	Whether to fall back to Yahoo when the paid provider fails. None = enabled for Yahoo, disabled for paid providers.

7.30 Search API keys

Several API keys for the `mf_search` / `websearch` tools can be stored in the config file. Environment variables always take precedence.

Field	Type	Default	Env var override	Description
<code>tavily_api_key</code>	<code>Option<String></code>	<code>None</code>	<code>TAVILY_API_KEY</code>	Tavily search API key for the <code>websearch</code> tool.
<code>langsearch_api_key</code>	<code>Option<String></code>	<code>None</code>	—	LangSearch API key for <code>mf_search</code> . Masked in diagnostics.
<code>perplexity_api_key</code>	<code>Option<String></code>	<code>None</code>	—	Perplexity Sonar API key for <code>mf_search</code> . Masked in diagnostics.
<code>exa_api_key</code>	<code>Option<String></code>	<code>None</code>	<code>EXA_API_KEY</code>	Exa Search API key for <code>mf_search</code> . Masked in diagnostics.
<code>serper_api_key</code>	<code>Option<String></code>	<code>None</code>	<code>SERPER_API_KEY</code>	Serper (Google Search) API key for <code>mf_search</code> . Masked in diagnostics.
<code>openalex_email</code>	<code>Option<String></code>	<code>None</code>	<code>OPENALEX_EMAIL</code>	OpenAlex polite-pool email for <code>mf_search</code> . Masked in diagnostics.

When present, `mf_search` queries the corresponding backend as an additional search engine alongside the keyless backends (OpenAlex, Wikipedia).

7.31 `sdd`

Spec-Driven Development (SDD) capability toggles. All flags default to `false` (opt-in).

```
{
  "sdd": {
    "clarification_markers": true,
    "quality_checklists": true,
    "constitution": true,
    "phase_minus_one_gates": true,
    "branch_per_spec": true,
    "research_artifacts": true,
    "data_model": true,
    "contracts": true,
    "quickstart": true,
    "test_first_ordering": true,
    "consistency_checks": true
  }
}
```

SddConfig schema

Field	Type	Default	Description
clarification_markers	bool	false	Enable [NEEDS CLARIFICATION] marker detection (FR-002).
quality_checklists	bool	false	Embed quality checklists in spec/plan templates (FR-006).
constitution	bool	false	Generate and parse CONSTITUTION.md (FR-007).
phase_minus_one_gate	bool	false	Enable Phase -1 pre-implementation gate validation (FR-008).
branch_per_spec	bool	false	Create a git branch per spec (FR-009).
research_artifacts	bool	false	Link research artifacts into SPEC.md frontmatter (FR-010).
data_model	bool	false	Generate data-model.md during /spec plan (FR-011).
contracts	bool	false	Generate contracts/ directory during /spec plan (FR-012).
quickstart	bool	false	Generate quickstart.md validation scenarios (FR-013).
test_first_ordering	bool	false	Enforce test-first file creation ordering in plans (FR-014).
consistency_checks	bool	false	Run ambiguity, contradiction, and gap consistency checks (FR-015).

Merge uses OR semantics: a flag enabled in either base or overlay stays enabled.

For the spec system, see docs/howtos/spec.md.

7.32 trigger

Dynamic trigger rule system configuration.

```
{
  "trigger": {
    "enabled": true,
    "poll_interval_secs": 30,
    "max_rules": 32
  }
}
```

TriggerConfig schema

Field	Type	Default	Description
enabled	bool	true	Master feature gate. When false, all trigger functionality no-ops.

Field	Type	Default	Description
poll_interval_sec	u32	30	Interval at which dynamic trigger rules poll their conditions.
max_rules	usize	32	Maximum number of dynamic trigger rules per session.

`is_empty()` returns `true` when all fields are at default values, so the section is omitted from serialised output when nothing is configured.

7.33 piegap

Pie feature gap toggles. Each flag gates a standalone pie-derived feature. All default to `false` (opt-in).

```
{
  "piegap": {
    "triggers": true,
    "mcp_notifications": true,
    "inbox": true,
    "hooks": true,
    "archive": true,
    "bug_report": true,
    "templates": true,
    "goal": true,
    "web_ui": true,
    "undo": true,
    "session_naming": true
  }
}
```

PieGapConfig schema

Field	Type	Default	Description
triggers	bool	false	Enable dynamic trigger rules (G-01).
mcp_notifications	bool	false	Enable MCP notification push events (G-02).
inbox	bool	false	Enable stateful loops + triage inbox (G-03).
hooks	bool	false	Enable lifecycle hooks (G-04).
archive	bool	false	Enable portable session archive export/import (G-05).
bug_report	bool	false	Enable bug report generation (G-06).
templates	bool	false	Enable reusable prompt templates (G-07).
goal	bool	false	Enable goal-based autonomous stop hook (G-10).
web_ui	bool	false	Enable browser-based web UI (G-12).

Field	Type	Default	Description
undo	bool	false	Enable /undo slash command (G-13).
session_naming	bool	false	Enable session naming (G-14).

Merge uses OR semantics. `is_empty()` returns true when no flags are enabled.

7.34 research

Research subsystem configuration.

```
{
  "research": {
    "open_access_recovery": true,
    "exclude_academic_engines": false,
    "contact_email": "user@example.com",
    "oa_min_full_text_chars": 1000,
    "max_concepts": 5,
    "max_findings": 20,
    "evaluate": {
      "enabled": true
    }
  }
}
```

ResearchConfig schema

Field	Type	Default	Description
open_access_recovery	bool	false	Enable open-access recovery via Unpaywall and Europe PMC for short scholarly sources (FR-011).
exclude_academic_engines	bool	false	Persistently exclude academically-classified search engines (OpenAlex) from research runs (spec researchnoacc FR-012). A per-run <code>--no-papers</code> takes precedence.
contact_email	Option<String>	None	Contact email required by Unpaywall's terms of service (FR-012).
oa_min_full_text_chars	usize	1000	Minimum full-text length (chars) that triggers OA recovery.
max_concepts	usize	5	Maximum number of concepts rendered in the report's <code>## Concepts</code> block (spec researchmax FR-008). A per-run <code>--max-concepts</code> takes precedence; 0 means unbounded.

Field	Type	Default	Description
max_findings	usize	20	Maximum number of findings rendered in the report's <code>## Findings</code> block (spec researchmax FR-008). A per-run <code>--max-findings</code> takes precedence; 0 means unbounded.
evaluate	ResearchEvaluation	<code>{"enabled": false}</code>	Self-evaluation scorecard settings (FR-015 of specs/opendeepresearch). When enabled, the research pipeline appends a deterministic quality scorecard (quality, relevance, groundedness, completeness, structure) to the report.

As of v1.0.77, `evaluate` can be configured in `ragent.json` and is omitted from serialized output when at its default value (`enabled: false`).

As of v1.0.100, the default web-phase timeout is 180 seconds and can be overridden per run with `--web-time N` (or `--web-phase-timeout-secs N`); 0 disables the deadline. When the deadline elapses, the run continues with the sources gathered so far. See `docs/howtos/research.md` for the full research workflow.

`open_access_recovery` uses OR semantics on merge. `contact_email` and `oa_min_full_text_chars` override base when present. `is_empty()` returns true when at default values.

`exclude_academic_engines` also uses OR semantics on merge and is omitted from serialized output when false. When enabled, research runs pass `exclude_engines = ["openalex"]` to `mf_search` so OpenAlex is never queried; a per-run `--no-papers` (alias `--no-scholarly`) takes precedence over the config value. See `docs/howtos/research.md`.

A per-run `--oa-enable` / `--no-oa` flag overrides `research.open_access_recovery`: the flag wins when set, otherwise the config value is used, and when neither is set recovery is off. See `docs/howtos/research.md`.

For the research system, see `docs/howtos/research.md`.

7.35 loop

Goal-driven /loop run budgets and checkpoints (see `docs/howtos/loopprogramming.md`). The whole section is omitted from serialized output when every field is at its default.

```
{
  "loop": {
    "max_steps": 512,
    "cost_limit": null,
    "error_retry_allowance": 3,
    "checkpoints": true,
    "checkpoint_timeout_secs": 120
  }
}
```

```
}
}
```

LoopConfig schema

Field	Type	Default	Description
max_steps	u32	512	Default step budget for a loop run (DEFAULT_LOOP_MAX_STEPS). A per-agent agent.<name>.max_steps override takes precedence.
cost_limit	Option<u64>	None	Token-cost budget (accumulated input + output) across a loop run; None means no gate.
error_retry_allowance	u32	3	Consecutive recoverable-failure tolerance before the loop stops with status error.
checkpoints	bool	true	Force destructive-action checkpoints before destructive tool calls even when an allow rule would auto-approve. Merge is monotonic: once disabled it stays disabled.
checkpoint_timeout_secs	u32	120	Seconds a forced checkpoint prompt waits for a reply before the timeout counts as denial.

Merge is sentinel-based: overlay fields equal to the compiled default keep the base (global) values; only fields the overlay configured away from the default win.

7.36 activity_log

```
{
  "activity_log": true
}
```

Field	Type	Default	Description
activity_log	bool	true	Record agent events to the activity-log SQLite store (append-only JSONL-style event log with RunId/EventId identifiers).

Toggle at runtime with /alog on|off.

8. Full Example File

The following is a comprehensive example showing every section. You do not need all of these — every section has defaults, so an empty {} is valid.

```
{
  "username": "alice",
  "defaultAgent": "coder",

  "provider": {
    "anthropic": {
      "env": ["ANTHROPIC_API_KEY"],
      "thinking": { "enabled": true, "level": "low" },
      "models": {
        "claude-sonnet-4-20250514": {
          "thinking": {
            "enabled": true,
            "level": "high",
            "budget_tokens": 16000
          }
        }
      }
    },
    "openai": {
      "env": ["OPENAI_API_KEY"]
    },
    "generic_openai": {
      "env": ["GENERIC_OPENAI_API_KEY"],
      "api": { "base_url": "http://127.0.0.1:8080" }
    },
    "router": {
      "enabled": true,
      "tiers": {
        "SIMPLE": {
          "models": [{ "provider": "ollama", "model": "llama3.2" }]
        },
        "MEDIUM": {
          "models": [
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
          ]
        },
        "COMPLEX": {
          "models": [
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
          ]
        },
        "REASONING": {
          "models": [{ "provider": "openai", "model": "o3" }]
        }
      }
    }
  }
}
```

```

    }
  },

  "agent": {
    "coder": {
      "model": "anthropic/claude-sonnet-4-20250514",
      "temperature": 0.2,
      "max_steps": 200,
      "skills": ["rust-review"]
    }
  },

  "permission": [
    { "permission": "file:write", "pattern": "src/**", "action": "allow" },
    { "permission": "file:write", "pattern": "secrets/**", "action": "deny" },
    { "permission": "bash", "pattern": "git push --force", "action": "deny" }
  ],

  "command": {
    "build": {
      "command": "cargo build --release",
      "description": "Build release binary"
    }
  },
  "mcp": {
    "filesystem": {
      "type": "stdio",
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-filesystem", "/home/user/docs"]
    }
  },

  "instructions": [
    "Always use rustfmt before writing Rust files."
  ],

  "skill_dirs": ["~/my-skills"],

  "experimental": {
    "parallel_tool_calls": false,
    "max_background_agents": 8,
    "background_agent_timeout": 3600
  },

  "bash": {
    "allowlist": ["curl", "wget"],
    "denylist": ["git push --force"]
  },

  "dirs": {

```

```

    "allowlist": ["src/**", "tests/**"],
    "denylist": ["secrets/**", ".env"]
  },

  "tool_visibility": {
    "office": true,
    "github": true,
    "gitlab": false,
    "teams": true,
    "agents": true,
    "plan": true,
    "codeindex": true,
    "masterfetch": true,
    "browser": true,
    "finance": true
  },

  "code_index": {
    "enabled": true,
    "max_file_size": 1048576,
    "extra_exclude_dirs": ["vendor"]
  },

  "stream": {
    "initial_response_timeout_secs": 300,
    "timeout_secs": 120,
    "max_retries": 4,
    "retry_backoff_secs": 2
  },

  "memory": {
    "enabled": true,
    "tier": "core",
    "structured": { "enabled": true },
    "semantic": {
      "enabled": false,
      "model": "all-MiniLM-L6-v2",
      "dimensions": 384
    },
    "retrieval": {
      "max_memories_per_prompt": 100,
      "max_memory_tokens": 4000,
      "recency_weight": 0.3,
      "relevance_weight": 0.7
    },
    "auto_extract": {
      "enabled": false,
      "require_confirmation": true
    }
  },

```

```

    "decay": {
      "factor": 0.95,
      "min_confidence": 0.1
    },
    "cross_project": {
      "enabled": false,
      "search_global": true,
      "project_override": true
    }
  },

  "compaction": {
    "auto": true,
    "threshold": 0.7,
    "buffer": 0.10,
    "keep": { "tokens": 0.20 }
  },

  "gitlab": {
    "instance_url": "https://gitlab.example.com",
    "token": "env:GITLAB_TOKEN",
    "username": "myuser"
  },

  "hidden_tools": [],

  "yolo": false,
  "edit_log": false,

  "prices": [
    {
      "model": "gpt-4o",
      "input_per_1m": 2.50,
      "output_per_1m": 10.00
    }
  ],

  "browser": {
    "cdp_endpoint": "http://127.0.0.1:9222",
    "default_headless": true
  },

  "channels": {
    "enabled": true,
    "telegram": {
      "bot_token": "env:TELEGRAM_BOT_TOKEN",
      "chat_id": "-100123"
    }
  }
},

```

```

"gmail": {
  "client_id": "env:GMAIL_CLIENT_ID",
  "client_secret": "env:GMAIL_CLIENT_SECRET"
},

"telemetry": {
  "otel": {
    "enabled": false,
    "endpoint": "http://localhost:4318",
    "protocol": "http",
    "export_interval_seconds": 30,
    "export_timeout_seconds": 10,
    "service_name": "ragent",
    "cardinality_limit": 1000
  }
},

"agent_perf": {
  "enabled": true,
  "step_budget_secs": 300,
  "stall_timeout_secs": 60,
  "max_concurrent_tools": 4,
  "parallel_independent_tools": true
},

"finance": {
  "provider": "yahoo",
  "min_call_interval_seconds": 5
},

"tavily_api_key": null,
"langsearch_api_key": null,
"perplexity_api_key": null,
"exa_api_key": null,
"serper_api_key": null,
"openalex_email": null,

"sdd": {
  "clarification_markers": false,
  "quality_checklists": false
},

"trigger": {
  "enabled": true,
  "poll_interval_secs": 30,
  "max_rules": 32
},

```



```

"piegap": {
  "triggers": false,
  "hooks": false,
  "undo": false
},

"research": {
  "open_access_recovery": false,
  "exclude_academic_engines": false,
  "contact_email": null,
  "oa_min_full_text_chars": 1000,
  "evaluate": {
    "enabled": false
  }
},

"loop": {
  "max_steps": 512,
  "cost_limit": null,
  "error_retry_allowance": 3,
  "checkpoints": true,
  "checkpoint_timeout_secs": 120
},

"activity_log": true
}

```

9. Common Recipes

Minimal config (just an API key reference)

```

{
  "provider": {
    "anthropic": { "env": ["ANTHROPIC_API_KEY"] }
  }
}

```

The API key itself is read from the environment variable; it does not need to be in the config file.

Local-first (Ollama only)

```

{
  "defaultAgent": "coder",
  "provider": {
    "ollama": {}
  }
}

```

ragent resolves to the first available local/self-hosted provider when no model is explicitly configured.

Custom OpenAI-compatible endpoint

```
{
  "provider": {
    "generic_openai": {
      "env": ["GENERIC_OPENAI_API_KEY"],
      "api": { "base_url": "http://127.0.0.1:8080" }
    }
  }
}
```

The base URL can also be set via `GENERIC_OPENAI_API_BASE` env var.

Allow writes to `src/`, deny secrets

```
{
  "permission": [
    { "permission": "file:write", "pattern": "src/**", "action": "allow" },
    { "permission": "file:write", "pattern": "secrets/**", "action": "deny" }
  ]
}
```

Enable GitHub and teams tools

```
{
  "tool_visibility": {
    "github": true,
    "teams": true
  }
}
```

Disable code indexing

```
{
  "code_index": { "enabled": false }
}
```

Semantic memory with embeddings

```
{
  "memory": {
    "semantic": {
      "enabled": true,
      "model": "all-MiniLM-L6-v2",
      "dimensions": 384
    }
  }
}
```

Requires the embeddings Cargo feature. When disabled, memory search falls back to FTS5-only mode.

OpenTelemetry export to a collector

```
{
  "telemetry": {
    "otel": {
      "enabled": true,
      "endpoint": "http://otel-collector:4318",
      "protocol": "http",
      "export_interval_seconds": 15,
      "service_name": "ragent-prod",
      "resource_attributes": {
        "deployment.environment": "production",
        "host.name": "build-server-01"
      }
    }
  }
}
```

Telegram notifications

```
{
  "channels": {
    "enabled": true,
    "telegram": {
      "bot_token": "env:TELEGRAM_BOT_TOKEN",
      "chat_id": "env:TELEGRAM_CHAT_ID"
    }
  }
}
```

Model Router with fallback chains

```
{
  "provider": {
    "router": {
      "enabled": true,
      "tiers": {
        "SIMPLE": {
          "models": [
            { "provider": "ollama", "model": "llama3.2" },
            { "provider": "openai", "model": "gpt-4o-mini" }
          ]
        },
        "MEDIUM": {
          "models": [
            { "provider": "anthropic", "model": "claude-sonnet-4-20250514" },
            { "provider": "openai", "model": "gpt-4o" }
          ]
        },
        "COMPLEX": {
```

```
    "models": [
      { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
    ],
    "REASONING": {
      "models": [
        { "provider": "openai", "model": "o3" },
        { "provider": "anthropic", "model": "claude-sonnet-4-20250514" }
      ]
    }
  }
}
```

Override pricing for a custom model

```
{
  "prices": [
    {
      "model": "my-custom-model",
      "input_per_1m": 1.00,
      "output_per_1m": 5.00
    }
  ]
}
```

Inline config via environment variable

```
RAGENT_CONFIG_CONTENT='{"defaultAgent":"ask","yolo":true}' ragent run "hello"
```

This is merged last, with the highest precedence.

10. Related Documents

Document	Topic
docs/howtos/tutorial.md	End-to-end tutorial from setup to release
docs/howtos/permissions.md	Permissions, autopilot, and YOLO mode
docs/howtos/tools.md	Complete tool reference
docs/howtos/tool-visibility.md	Tool family visibility toggles
docs/howtos/custom-agents.md	Custom agent definitions (OASF / Markdown)
docs/howtos/teams.md	Teams and swarm coordination
docs/howtos/hooks.md	Lifecycle hooks
docs/howtos/codeindex.md	Code index (tree-sitter, search, graph)

Document	Topic
<code>docs/howtos/spec.md</code>	Spec lifecycle management
<code>docs/howtos/research.md</code>	Research system
<code>docs/howtos/finance.md</code>	Stock and currency tools
<code>docs/howtos/communications.md</code>	Gmail and messaging channels
<code>docs/howtos/reverse.md</code>	GitHub repo reverse-engineering
<code>README.md</code>	Project overview and quick start
<code>SPEC.md</code>	Full configuration schema reference
<code>QUICKSTART.md</code>	Quick start guide